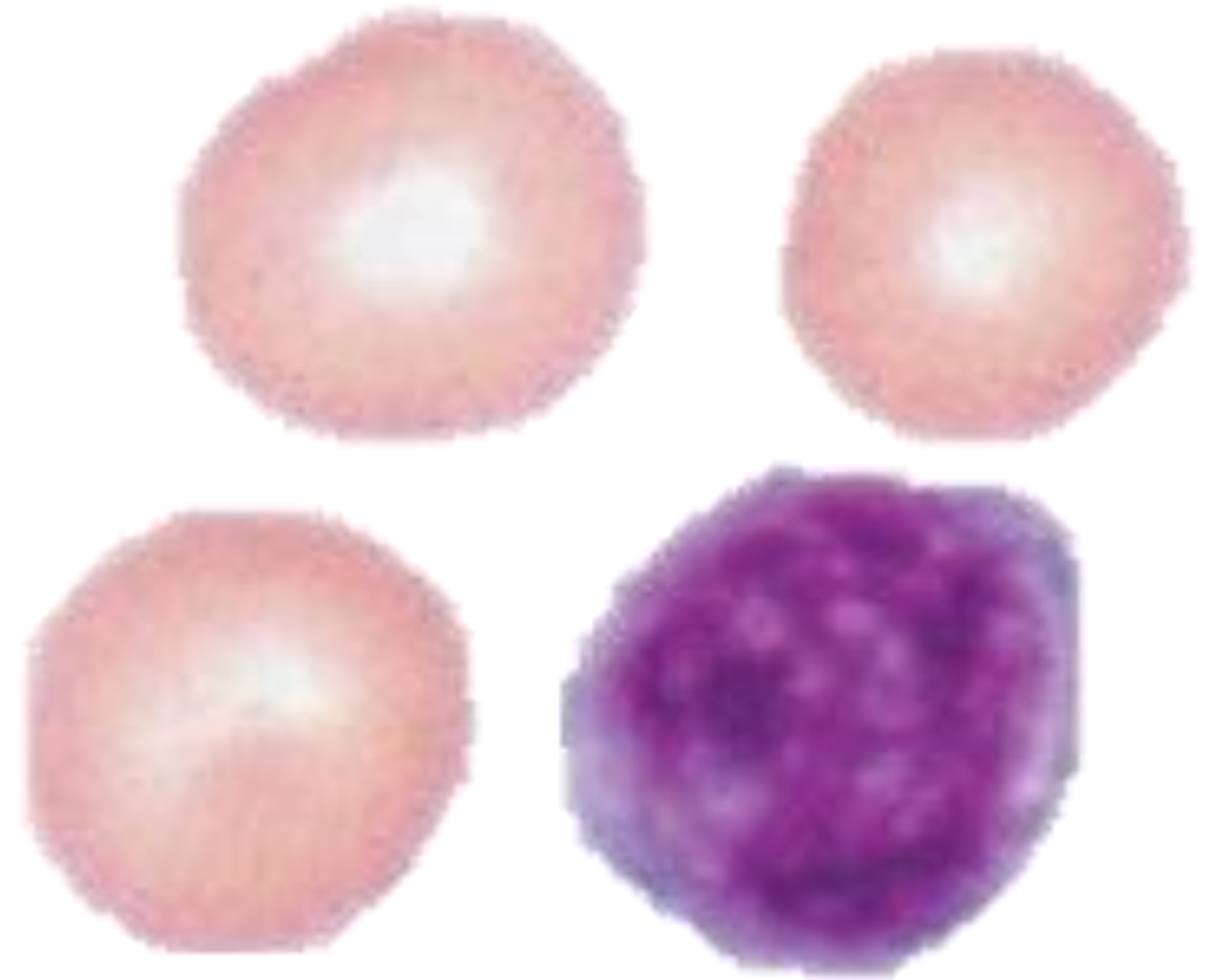


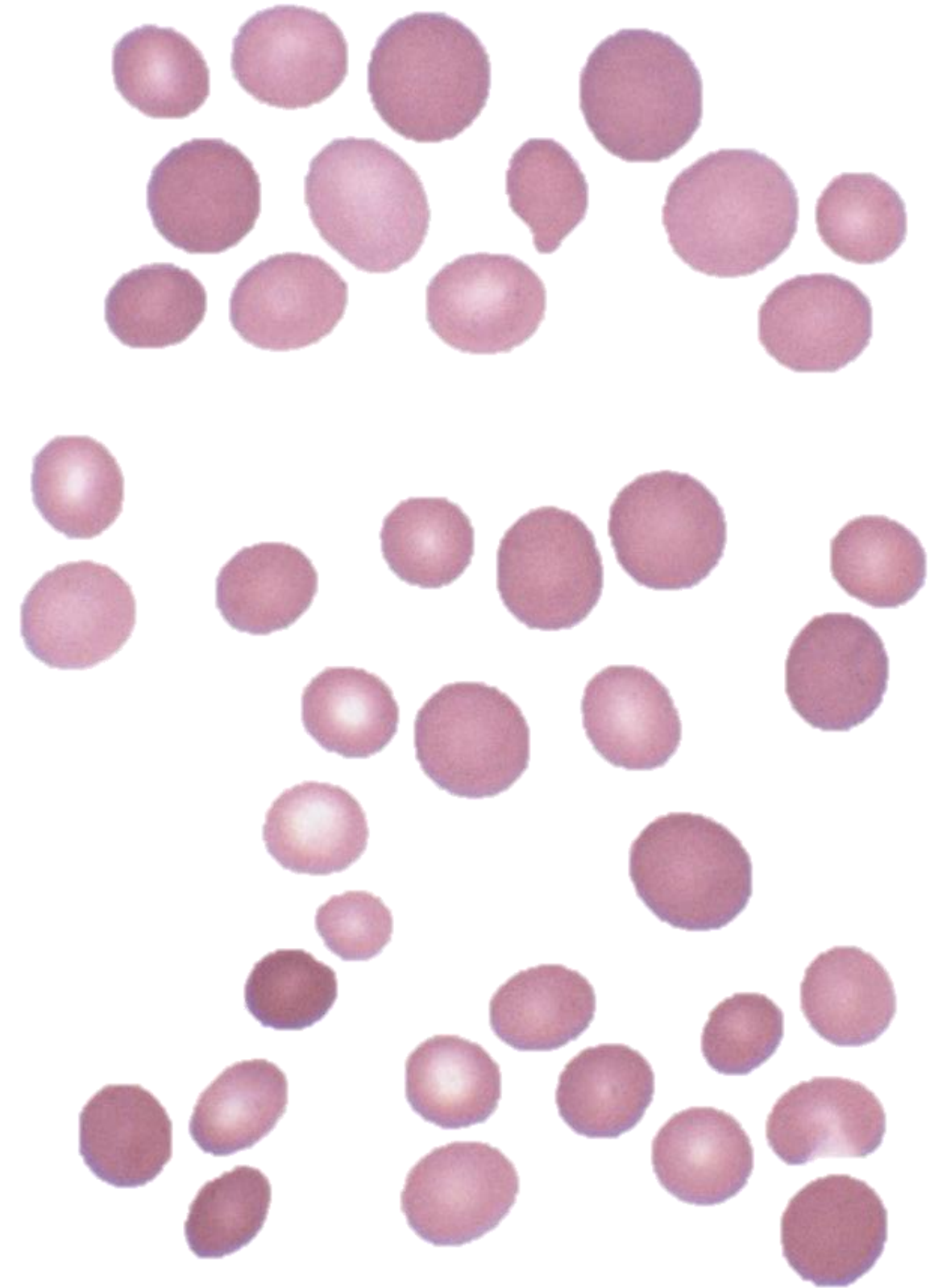
Erythrocyte / RBC

- Discocyte due to biconcave shape
- Size 7–8 μm ; MCV 80–100 fL
 - Used for micro/macro/aniso
- Central area of pallor surrounded/caused by a rim of pink-staining HGB
 - $\sim 1/3$ diameter of RBC
 - Used for hypochromasia



Anisocytosis

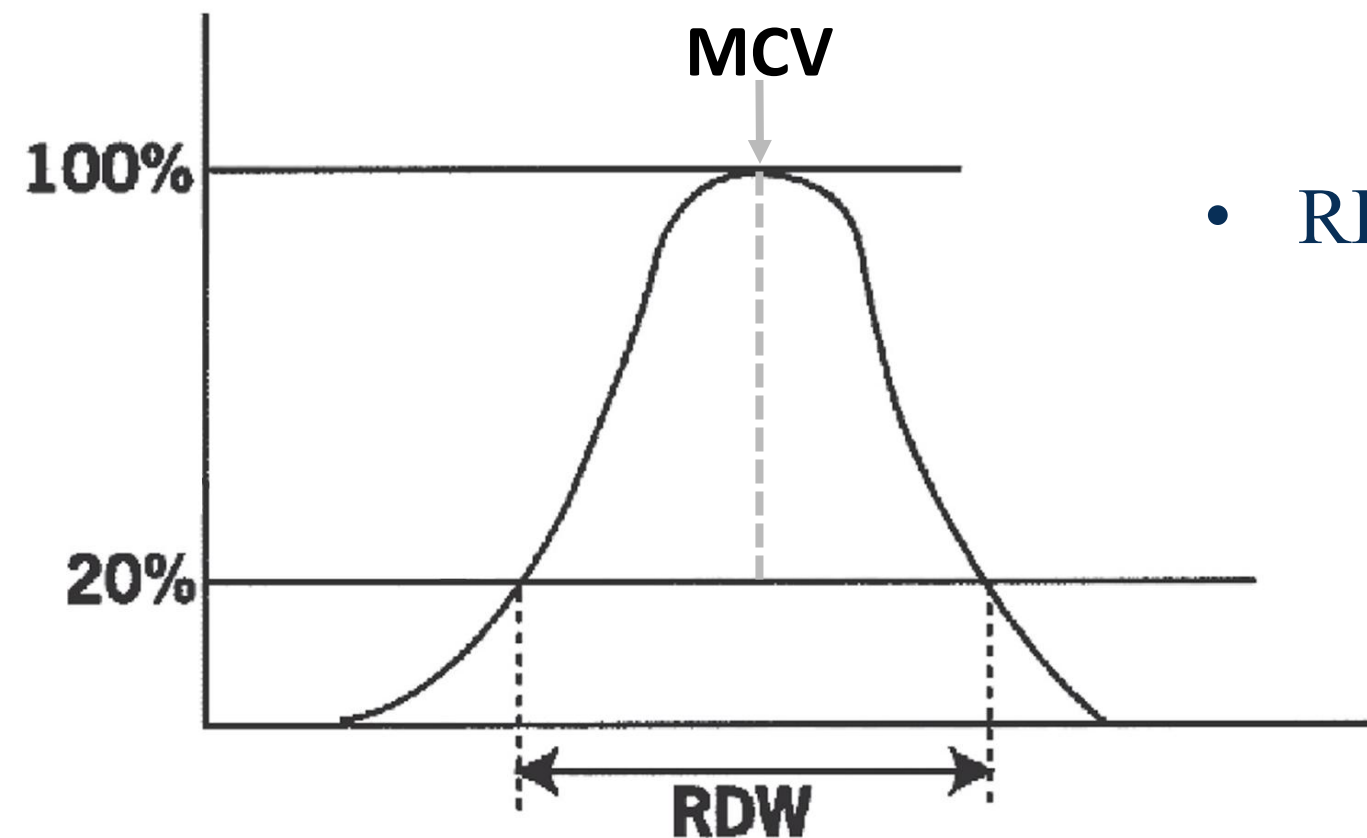
- Variation in red cell size
 - Detected by reviewing RDW-CV and MCV
 - Confirmed by reviewing smear
- RBC sizes described as
 - $MCV < 80 \text{ fL}$ = Microcytic
 - $MCV \geq 80 \text{ fL}$ and $\leq 100 \text{ fL}$ = Normocytic
 - $MCV > 100 \text{ fL}$ = Macrocytic



Anisocytosis

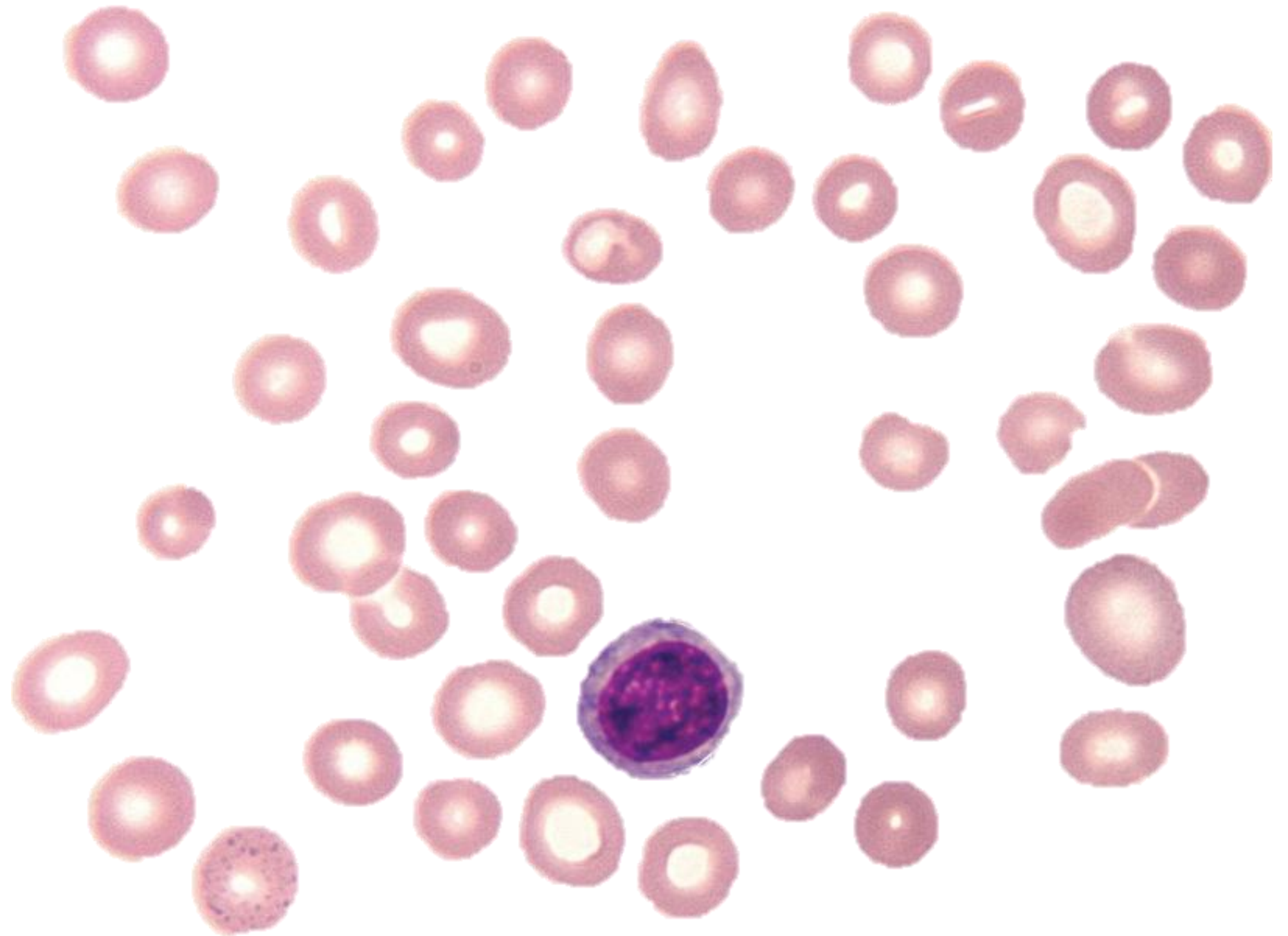
- $\text{RDW-CV} > 14.6 \%$
 - Suggests different RBC sizes present

- $\text{RDW-CV} = (\text{RBC Volume SD} / \text{MCV}) * 100$



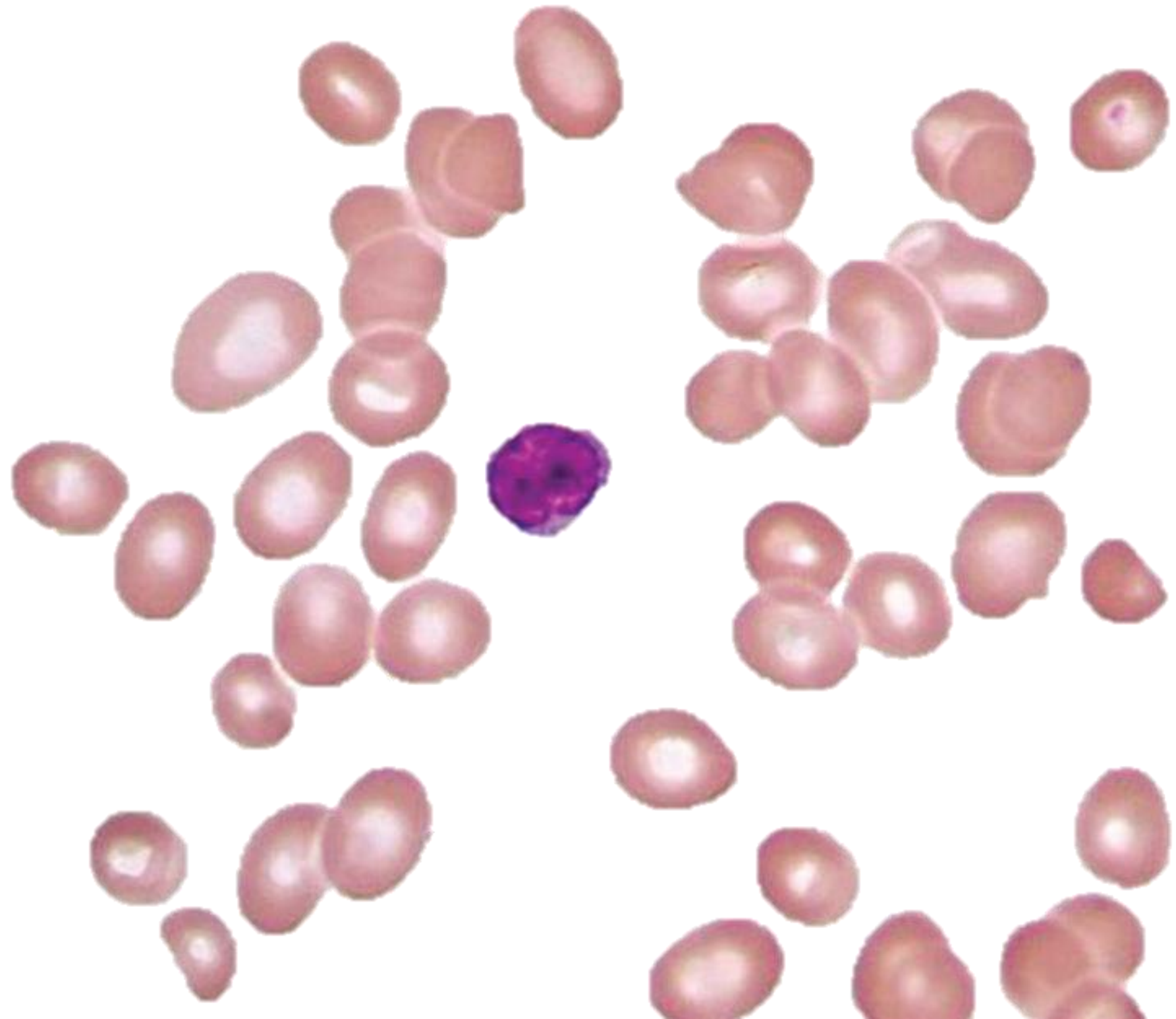
Microcytes

- RBCs
 - Diameter $< 7.0 \mu\text{m}$
 - MCV $< 80 \text{ fL}$
 - Usually hypochromic [by MCH] but can be normochromic
- Microcytes in the shape of spheres
 - Microspherocytes
 - Can appear “hyperchromic”



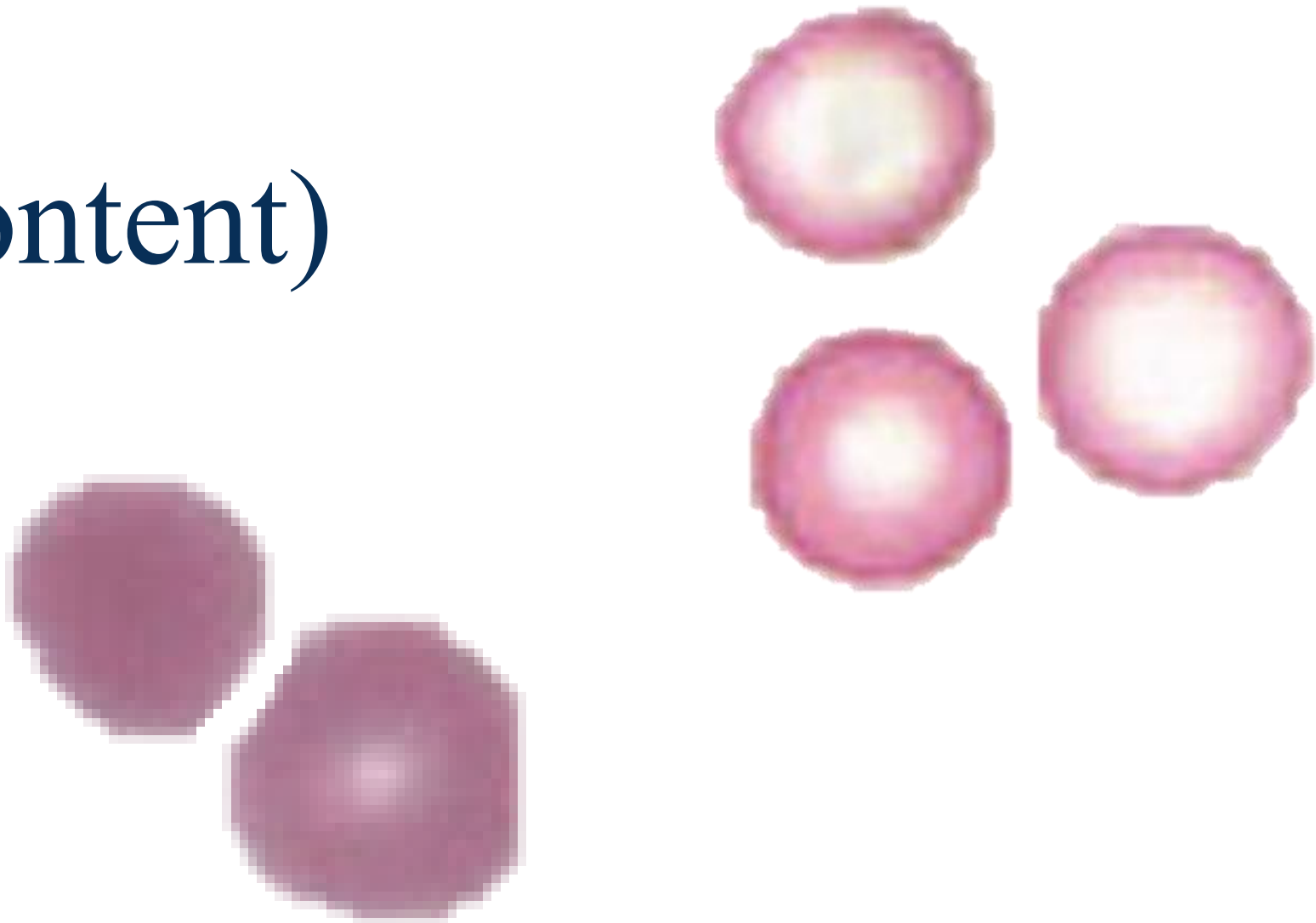
Macrocytes

- RBCs
 - Diameter $> 8.0 \mu\text{m}$
 - $\text{MCV} > 100 \text{ fL}$
 - Usually contains an adequate amount of Hgb
 - Normal MCHC
 - Normal to $\uparrow$ MCH



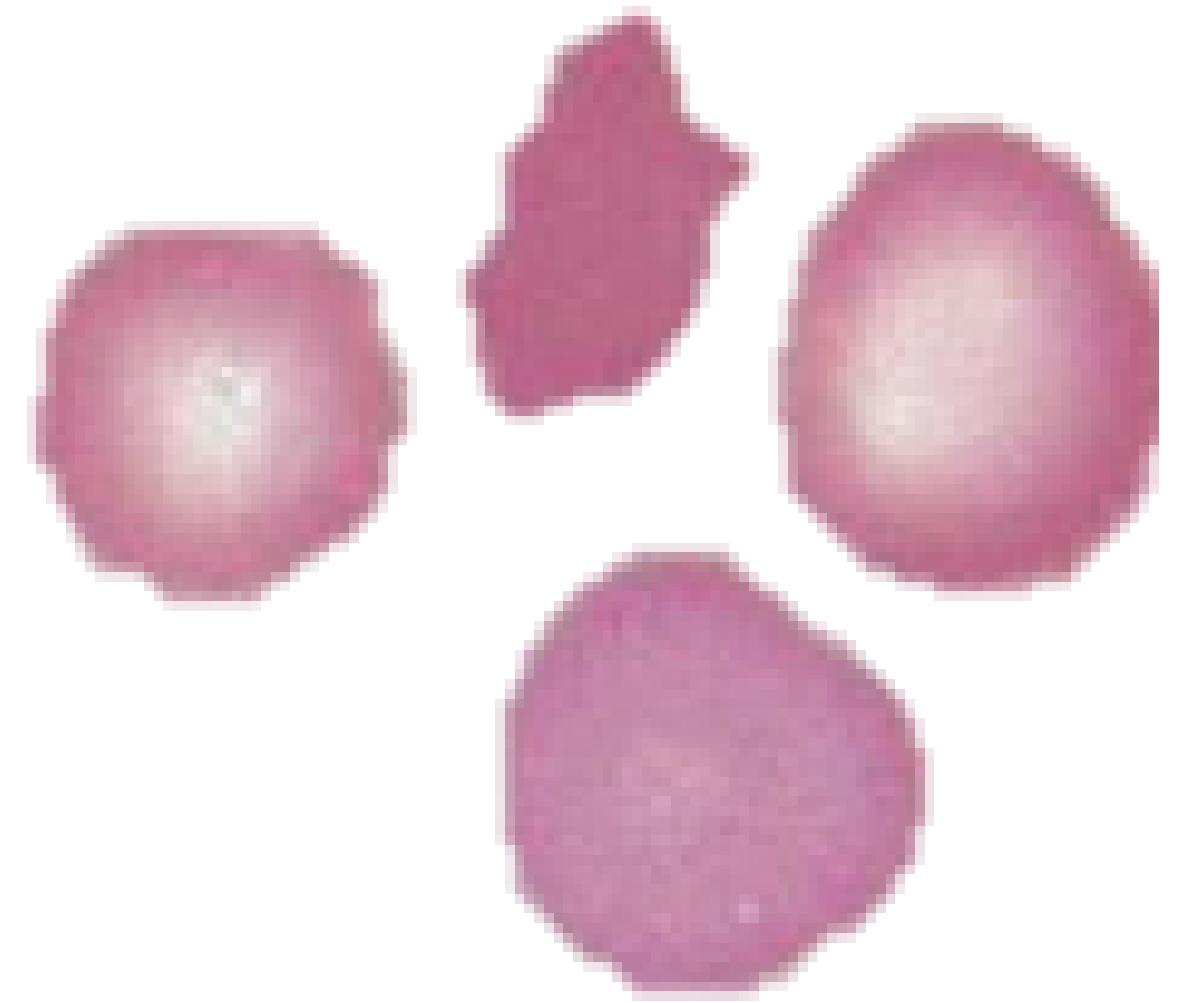
[Hypo]chromasia (HGB content)

- Normochromic (~1/3 central pallor)
 - MCH = 28 - 34 pg | MCHC 32 – 36 g/dL
- Hypochromic (> 1/3 central pallor)
 - MCH < 28 pg | MCHC < 32 g/dL
- “Hyperchromic” (< 1/3 central pallor)
 - MCH > 34 pg | MCHC > 36 g/dL



Terminology	Description	Associated Physiological or Disease States
Hypochromia	Decreased concentration of hemoglobin in the red cell Red cells have an increased area of central pallor (> 1/3 diameter of cell)	May be present in iron deficiency anemia, thalassemia, and other anemias associated with a defect in hemoglobin production (Chapter 12–14)

[Poly]chromasia (RNA content)



- Reticulocytes (Polychromatophilic RBCs)
 - Usually larger than normal RBCs
 - Bluish tinge on Romanowsky/Wright stain
 - Caused by the residual RNA in the cytoplasm
 - ↑ numbers erythroid hyperplasia
 - eg ↓ RBC survival, hemorrhage

Terminology	Description	Associated Physiological or Disease States
Polychromasia	Young red cells containing residual RNA Stain a pinkish-gray to pinkish-blue color on Wright's stained blood smears Usually appear slightly larger than mature red cells	Found in increased numbers in hemolytic anemias, newborns, recovery from acute hemorrhage (Chapter 16–20)